MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

MANIPAL UNIVERSITY JAIPUR

Department of AIML

Course Hand-out

A. Basic Details:

Programme Name:	B. Tech CSE (AIML)
Course Name:	Project Based Learning-3
Course Code:	AIM3170
LTPC *(Lecture Tutorial Practical Credits):*	1
Session:	Jul 2026- Nov 2026
Class:	2024-2028
Course Coordinator:	Dr. Pratima Kumari, Mrs. Babita Kinha
Additional Practitioner(s) – if any *(Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.):*	To be identified and appointed later/ or at the beginning of the semester

B. Introduction: This course aims at developing innovative skills in the students whereby they apply in totality the knowledge and skills gained through the course work in the solution of problem or by undertaking a project. Project work, therefore, should match the strengths of students. The project assignment can be individual assignment or a group assignment. The project work identified in collaboration with industry should be preferred. Each teacher is expected to guide the project work of 5-6 students. The project assignments may consist of: Programming customer-based applications, Web page designing (Only dynamic), Database applications, Software Development etc. Execution through mentor mentee policy.

C. Course Outcomes: At the end of the course, students will be able to

CO Statement	CO	Level	Target Attainment %	Target Attainment Level
Apply engineering principles and tools to design and develop project deliverables effectively.	AIM3170.1	Apply (Level 3)	85%	Level 3
Analyse project challenges and propose feasible solutions using systematic methodologies.	AIM3170.2	Analyse (Level 4)	80%	Level 3

Evaluate the performance and sustainability of project outcomes with technical and ethical considerations.	AIM3170.3	Evaluate (Level 5)	75%	Level 2
Create innovative and impactful solutions, demonstrating teamwork, creativity, and communication skills.	AIM3170.4	Create (Level 6)	75%	Level 2

Information about attainment levels:

Attainment (%)	Level
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes:

The Program Outcomes will be as follows-

PO 1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

PO 9: Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PSO 1: Graduates will be able to examine the applications of Artificial Intelligence and Machine Learning in real-life problems
PSO 2: Graduates will be able to design and implement intelligent systems for multidisciplinary problems

E. Assessment Plan: Clearly write the criteria, its description, and associated marks for assessment of student achievements. In addition, the attendance requirements, assignments need to be mentioned in this section.

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Updated Synopsis submission	**10**
	Weekly Progress Report	**15**
	Analysis process ppt	**15**
	Design and Implement status in documentation	**20**
End Term Exam (Summative)	End Term Exam (Final report with plagiarism and Presentation)	**40**
	Total	**100**

F. Syllabus: This course aims at developing innovative skills in the students whereby they apply in totality the knowledge and skills gained through the course work in the solution of problem or by undertaking a project. Project work, therefore, should match the strengths of students. The project assignment can be individual assignment or a group assignment. The project work identified in collaboration with industry should be preferred. Each teacher is expected to guide the project work of 5-6 students. The project assignments may consist of: Programming customer-based applications, Web page designing (Only dynamic), Database applications, Software Development etc. Execution through mentor mentee policy.

G. Course Articulation Matrix: Couse articulation indicates the mapping of CO statement(s) to some PO/ PSO statement(s). Here value 0 indicates no correlation, 1 as low correlation, 2 as moderate correlation, and 3 as substantial / strong correlation.

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES													
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2

AI3170.1	Apply engineering principles and tools to design and develop project deliverables effectively.	2	2	1	2	3	2	2	2	1		1	2	2	1
AI3170.2	Analyze project challenges and propose feasible solutions using systematic methodologies.	2		3		1	2	1	2	3	2	2	3	1	1
AI3170.3	Evaluate the performance and sustainability of project outcomes with technical and ethical considerations.	1		1		3			2	2	3		3	1	
AI3170.4	Create innovative and impactful solutions, demonstrating teamwork, creativity, and communication skills.	1				3			2	1	3		3	1	

Note:

- Remember (Level 1).
- Understand (Level 2).
- Apply (Level 3).
- Analyse (Level 4).
- Evaluate (Level 5).
- Create (Level 6).

Course Coordinator

Name: Dr. Pratima Kumari

MUJID: MUJ2330

Head of the Department

Student Representative

Name:

Registration No:

Name: Mrs. Babita Kinha

MUJID: MUJ2322